Residual Adaptive Blending Units for Physics informed Neuronal Networks

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Maestría en Matemáticas Aplicadas

Convection Equation

Se introduce *Adaptative Blending Units*:

$$f(x) = \sum_{i=1}^{N} G(\alpha_i)\sigma_i(\beta_i x).$$

Se considera la ecuación de convección en lineal en 1D:

$$\begin{aligned} u_t + \beta u_x &= 0, \quad x \in [0, 2\pi], \quad t \in [0,1], \\ u(0,x) &= \sin(x), \\ u(t,0) &= u(t, 2\pi). \end{aligned}$$

Donde $\beta = 64$. Usaron puntos de colocación fijos de $N_r = 6400$, $N_{ic} = 512$ y $N_{bc} = 200$. La arquitectura es una fully-connected de 6 capas con 64 neuronas cada. Se dejo correr Adam por $100{,}000$ epocas.

En la replicación se usó la misma arquitectura, los mismos puntos de colocación, el mismo optimizador. Se usaron funciones de activación $\sigma = \{\sin, \cos, \text{racional}, \tanh\}$. Se incluye algoritmos LRA, remuestreo estocástico en cada época y tasa de aprendizaje con decaimiento de coseno. Se corrio por $40{,}000$ épocas con Adam como aproximación.

$$\sigma(x) = G(\alpha_1)\left(\frac{1}{1+x^2}\right) + G(\alpha_2)(\cos(\omega_0 x)) + G(\alpha_3)\sin(\omega_0 x) + G(\alpha_4)\tanh(x)$$

Algorithm 1: Learning rate annealing for physics-informed neural networks

Consider a physics-informed neural network $f_\theta(\boldsymbol{x})$ with parameters θ and a loss function

$$\mathcal{L}(\theta) := \mathcal{L}_r(\theta) + \sum_{i=1}^{M} \lambda_i \mathcal{L}_i(\theta),$$

where $\mathcal{L}_r(\theta)$ denotes the PDE residual loss, the $\mathcal{L}_i(\theta)$ correspond to data-fit terms (e.g., measurements, initial or boundary conditions, etc.), and $\lambda_i = 1$, $i = 1, \ldots, M$ are free parameters used to balance the interplay between the different loss terms. Then use S steps of a gradient descent algorithm to update the parameters θ as:

for $n = 1, \ldots, S$ **do**

(a) Compute $\hat{\lambda}_i$ by

$$\hat{\lambda}_i = \frac{\max_\theta\{|\nabla_\theta \mathcal{L}_r(\theta_n)|\}}{\overline{|\nabla_\theta \mathcal{L}_i(\theta_n)|}}, \quad i = 1, \ldots, M, \tag{40}$$

where $\overline{|\nabla_\theta \mathcal{L}_i(\theta_n)|}$ denotes the mean of $|\nabla_\theta \mathcal{L}_i(\theta_n)|$ with respect to parameters θ.

(b) Update the weights λ_i using a moving average of the form

$$\lambda_i = (1-\alpha)\lambda_i + \alpha\hat{\lambda}_i, \quad i = 1, \ldots, M. \tag{41}$$

(c) Update the parameters θ via gradient descent

$$\theta_{n+1} = \theta_n - \eta\nabla_\theta \mathcal{L}_r(\theta_n) - \eta\sum_{i=1}^{M} \lambda_i \nabla_\theta \mathcal{L}_i(\theta_n) \tag{42}$$

end

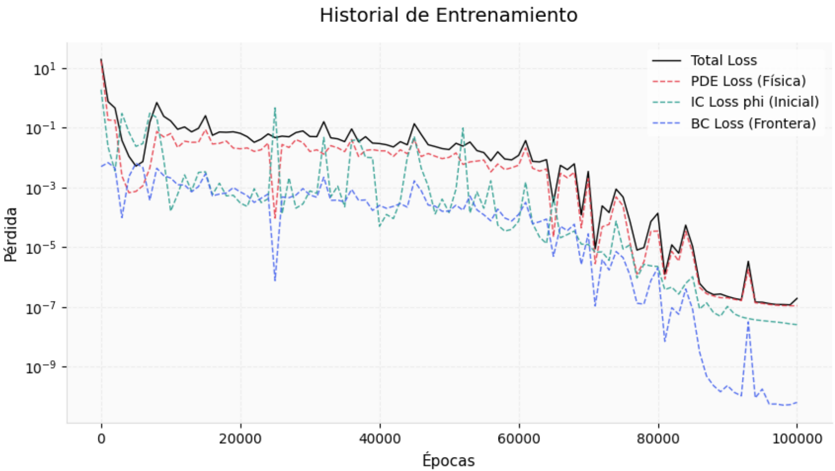
The recommended hyper-parameter values are: $\eta = 10^{-3}$ and $\alpha = 0.9$.

Resultados publicados

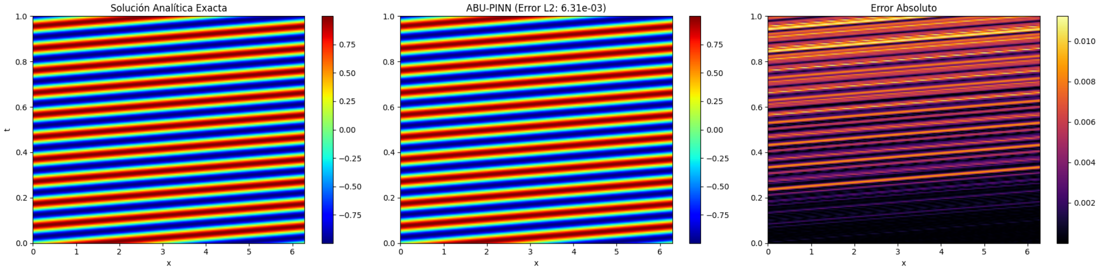
Table 5: Comparisons of standard activation functions, adaptive activation functions and their counterparts with adaptive slopes (AS) on 1D time-dependent PDEs. We report the L_2 relative error (%) for each problem and the average error rate over all problems. The better results are **bold-faced**.

Method	Convection equation	Burgers' equation	Allen-Cahn equation	KdV equation	Cahn-Hilliard equation	Average error
sin	0.36±0.15	5.18±3.73	3.57±0.64	0.63±0.17	1.72±0.61	2.29
tanh	6.83±4.79	0.26±0.13	1.34±0.54	1.32±1.12	4.02±4.56	2.75
sigmoid	70.38±2.98	1.24±1.05	1.63±0.13	2.34±0.53	3.12±2.72	15.74
GELU	39.29±35.51	4.43±2.87	3.93±0.78	1.21±0.41	1.01±1.27	9.97
Swish	5.59±2.18	8.27±4.35	5.56±1.28	1.73±0.10	2.22±2.60	4.67
Softplus	55.39±2.46	17.75±8.11	17.72±6.04	6.23±0.44	9.67±2.57	21.35
ELU	6.67±0.96	46.34±2.36	52.55±2.95	78.95±2.57	90.77±2.07	55.66
SLAF	0.36±0.18	43.93±0.66	33.93±9.97	25.23±1.28	52.57±27.93	31.20
PAU	45.78±35.47	48.31±8.21	43.83±15.18	68.11±13.49	115.59±3.79	64.32
ACON	3.55±1.66	1.18±1.55	3.88±1.82	1.52±0.35	2.46±1.96	2.52
ABU-PINN	**0.06±0.03**	**0.21±0.11**	**0.76±0.26**	**0.34±0.05**	**0.50±0.15**	**0.37**
With AS						
sin	0.28±0.08	1.82±1.58	3.57±0.46	0.57±0.19	1.73±0.73	1.59
tanh	3.06±1.65	0.16±0.09	0.81±0.21	1.71±1.53	2.11±0.49	1.57
GELU	38.00±39.53	0.71±0.38	1.99±0.63	0.79±0.17	0.96±0.57	8.49
ABU-PINN	**0.05±0.02**	**0.15±0.10**	**0.58±0.15**	**0.34±0.08**	**0.35±0.07**	**0.29**

Función de costo



Representabilidad y Error



Cahn-Hilliard Equation

Se considera la ecuación de Cahn-Hilliard:

$$u_t - \nabla^2 \left(-\lambda_1\lambda_2 h + \lambda_2(u^3 - u)\right) = 0, \quad h = \nabla^2 u, \quad x \in [-1,1], \quad t \in [0,1], \tag{1}$$

$$u(0,x) = \cos(\pi x) - \exp(-4(\pi x)^2), \tag{2}$$

$$u(t,-1) = u(t,1), \quad u_x(t,-1) = u_x(t,1), \tag{3}$$

$$h(t,-1) = h(t,1), \quad h_x(t,-1) = h_x(t,1), \tag{4}$$

Dónde $\lambda_1 = 0{,}02$ y $\lambda_2 = 1$. Usaron puntos de colocación fijos de $N_r = 15000$, $N_{ic} = 256$ y $N_{bc} = 100$. La arquitectura es una fully-connected de 4 capas con 32 neuronas cada. Se dejo correr Adam por $100{,}000$ epocas y $15{,}000$ épocas con L-BFGS. Se usaron los mismos algortimos que en ecuación de convección.

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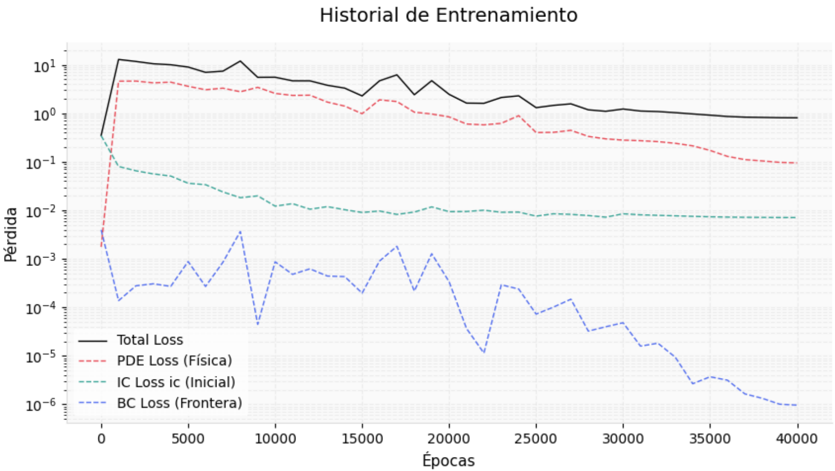

Representabilidad y Error